

Drone Mapper — Assignment 3 High-Level Design

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TAU Advanced Topics in Programming (2026B). Assignment 3 splits the ex2 monolithic simulator into three separately built projects: a `simulator_207190406_209543255` executable that `dlopen`s `Algorithm_207190406_209543255.so` and `MissionControl_207190406_209543255.so`, then runs many missions in parallel under `-comparative` or `-competition` mode.

Artifact	Name
Executable	<code>simulator_207190406_209543255</code>
Algorithm plugin	<code>Algorithm_207190406_209543255.so</code>
MissionControl plugin	<code>MissionControl_207190406_209543255.so</code>

Plugin / UserCommon **namespaces** (code): `algorithm_207190406_209543255`, `mission_control_207190406_209543255`, `user_common_207190406_209543255`.

Folder responsibilities

Folder	Role	Build artifact
<code>common/</code>	Course-published interfaces used by more than one project. Read-only — we do not modify it.	none (INTERFACE lib <code>common::common</code>)
<code>Simulator/common_simulator/include/Simulator/</code>	Course-published interfaces used only by the Simulator	none
<code>Simulator/{include,src}/Simulator</code>	Course-published implementation	<code>simulator_207190406_209543255</code> executable
<code>MissionControl/common_mission_control/include/MissionControl/</code>	Course-published <code>IDroneControl</code>	none

Folder	Role	Build artifact
MissionControl/{include,src}/	Our mission control implementation	MissionControl_207190406_209543255.so
Algorithm/{include,src}/	Our mapping algorithm	Algorithm_207190406_209543255.so
UserCommon/	Our code needed by more than one project. No build file — sources compile into each consumer.	none

The course staff placement rule: mocks and world simulation live in `Simulator/` because they would be replaced by real hardware APIs in production; mission control and the mapping algorithm live in their respective plugin projects. Shared cross-boundary interfaces stay in `common/`; simulator-only interfaces stay in `Simulator/common_simulator/`.

Main components

- **main (Simulator/src/main.cpp)** — Parses CLI via `parseSimulationCliArgs (SimulationCliArgs)`, creates the timestamped output directory with `createOutputDir`, loads the composition YAML, loads plugins through `PluginLoader`, builds one `SimulationRunFactoryImpl` per plugin binding, invokes `runPluginMatrix(bindings, composition, output_root, num_threads)` (`expandRunMatrix` runs inside that call), writes per-plugin `writeSimulationOutputYaml` then `writeModeReport` → `writeComparativeReport` / `writeCompetitiveReport`, then destroys plugin objects and calls `PluginLoader::unloadAll()` before return.
- **SimulationCli / SimulatorPaths (Simulator/io/SimulatorPaths.h)** — `parseSimulationCliArgs` returns `SimulationCliArgs`. Path helpers: `createOutputDir` (fresh `comparative_results_<UTC>` / `competition_<UTC>` folder) and `errorLogPathFromOutputMap` (derive `<plugin>_run_NNNN_error.log`).
- **YamlConfigParsers (Simulator/io/YamlConfigParsers.h)** — Five parsers: `parseCompositionFile`, `parseSimulationConfig`, `parseMissionConfig`, `parseDroneConfig`, `parseLidarConfig`. Parse failures are logged immediately through `RunErrorLog` / `IRunErrorLog`. A non-scalar `simulation_config` or a failed nested drone/lidar parse makes

`parseCompositionFile` return `ok = false` (`COMPOSITION_INVALID`) instead of throwing or keeping a default-constructed config.

- **SimulatorReports** (`Simulator/io/SimulatorReports.h`) — `writeComparativeReport`, `writeCompetitiveReport`, `writeSimulationOutputYaml`. `writeModeReport` in `main.cpp` picks the mode writer.
- **PluginLoader** — `dlopen`s each `.so` once on the main thread (`loadAlgorithmSo`, `loadMissionControlsFromDirectory`, or the competitive-mode equivalents). Handles are `DlHandle` (`unique_ptr<void, DlCloser>`). After each successful load it takes the factory registered by the plugin's static constructor into a `LoadedAlgorithmPlugin` / `LoadedMissionControlPlugin`. Access is `algorithmCount` / `algorithmAt` (and the mission-control equivalents) — no `algorithms()` vector getter. Never reloads a path already held open. `unloadAll()` drops factories then `dlclose`s every handle.
- **PluginRegistrar** — Simulator-owned singleton. `MappingAlgorithmRegistration` / `MissionControlRegistration` objects (defined in registration headers, `.cpp` in Simulator only) call `setPendingAlgorithmFactory` / `setPendingMissionControlFactory` during `dlopen`; the loader immediately `takePending*Factory()` so each registration is consumed exactly once. Factory aliases are `MappingAlgorithmFactory` / `MissionControlFactory`.
- **runPluginMatrix** — Free-function module (`RunMatrixOrchestrator.h`). Expands a composition into a flat `MatrixCell` list via `expandRunMatrix` (`simulation × mission × drone × lidar`) **inside** `runPluginMatrix(bindings, composition, output_root, num_threads)`, then runs every cell for every plugin binding. Pre-sizes the result table; each slot is written exactly once.
- **distributeWork** — Free-function scheduler. `num_threads` absent or 1 → main thread only. `N >= 2` → up to N worker threads (capped at cell count); main joins. Wraps per-cell work in `try/catch` so a plugin throw cannot terminate the process.
- **SimulationRunFactoryImpl** — Implements `simulator::ISimulationRunFactory`. Loads hidden and output `Map3DImpl` instances, constructs `MockGPS`, `MockLidar`, `MockMovement`, creates fresh plugin instances from the loaded factories, wires `MissionControlDependencies` / `MappingAlgorithmDependencies`, and returns a `SimulationRunImpl`.
- **SimulationRunImpl** — Implements `simulator::ISimulationRun`. Calls `missionControl->runMission()`, saves the output map, scores with `compareMaps(hidden, output, spawn)` against the hidden map, and returns `SimulationResult`. Per-run `RunErrorLog` path comes from `errorLogPathFromOutputMap`. Contains exceptions from `runMission()` so a single bad run scores `kErrorScore` (-1) without aborting the matrix.

- **Mocks + maps + scoring (Simulator/src/)** — `Map3DImpl` (hidden + output), `MockGPS`, `MockLidar`, `MockMovement` (holds the hidden map; throws on wall/boundary collision), `compareMaps`. `makeMap3D` is src-only (`Map3DNpy.h`).
- **MissionControlImpl_207190406_209543255** — Plugin entry point implementing `common::IMissionControl`. Creates `DroneControlImpl`, loops `step()` until the mission finishes or hits max steps, returns `MissionRunResult`.
- **DroneControlImpl** — Implements `mission_control::IDroneControl`. Each step carves the drone footprint (`markDroneFootprintEmpty`), then invokes `nextStep` once unless draining a split-oversize queue (`pending_movements_`). Fragment steps skip `nextStep` and skip an extra scan. `applyScanIfRequested` still runs after a recoverable `Continue` when the original command carries `scan_orientation`. Finished algorithm status skips move/scan. Oversize Advance/Elevate/Rotate is split into in-limit fragments. Unsupported types are retried then thrown (CI3), not returned as `Error`.
- **MappingAlgorithmImpl_207190406_209543255** — Plugin entry point implementing `common::IMappingAlgorithm`. Reads the world through `const common::IMap3D&` only. Uses `WavefrontPlanner`, which composes `MappingAlgorithmFrontier` and calls `PathShaping` / `ScanPlanning` to produce an `ExplorationPlan`, then emits movement plus a score-aware scan toward that cluster. Scan templates are `ConeTemplateCache` / `VoxelStamp`.
- **SimulationCoordUtil** — Shared world/voxel helpers: `worldInitialDronePosition`, `forEachSphereSample`.
- **SimulationImpl** — Implements published `simulator::ISimulation`. Constructor takes one `ISimulationRunFactory&`, the plugin filename used in per-run map names, and `num_threads`. `run(composition, output_path)` delegates to `runPluginMatrix` for that single binding and returns `SimulationManagerReport`. Comparative/competitive CLI, plugin folders, and aggregate YAML still live in `main` — `ISimulation::run` cannot express mode or a plugin set.

Class diagram

```
classDiagram
    direction TB
    class main {
        +writeModeReport()
    }
```

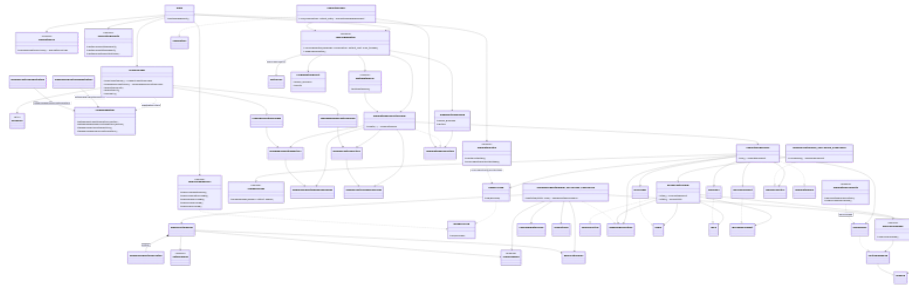


Figure 1: Class overview

```

class SimulationCli {
  <<module>>
  +parseSimulationCliArgs() SimulationCliArgs
}

class SimulatorPaths {
  <<module>>
  +createOutputDir()
  +errorLogPathFromOutputMap()
}

class YamlConfigParsers {
  <<module>>
  +parseCompositionFile()
  +parseSimulationConfig()
  +parseMissionConfig()
  +parseDroneConfig()
  +parseLidarConfig()
}

class SimulatorReports {
  <<module>>
  +writeComparativeReport()
  +writeCompetitiveReport()
  +writeSimulationOutputYaml()
}

class runPluginMatrix {
  <<module>>
  +runPluginMatrix(bindings, composition, output_root, num_threads)
  +expandRunMatrix()
}

```

```

class distributeWork {
    <<module>>
    +distributeWork()
}

class compareMaps {
    <<module>>
    +compareMaps(hidden, output, spawn)
}

class applyScanToMap {
    <<module>>
    +applyScanToMap()
}

class PluginLoader {
    +loadAlgorithmSo() LoadedAlgorithmPlugin
    +loadMissionControlSo() LoadedMissionControlPlugin
    +algorithmCount()
    +algorithmAt()
    +unloadAll()
}

class DlHandle {
    <<RAII>>
}

class PluginRegistrar {
    +setPendingAlgorithmFactory(factory)
    +setPendingMissionControlFactory(factory)
    +takePendingAlgorithmFactory()
    +takePendingMissionControlFactory()
}

class MappingAlgorithmRegistration
class MissionControlRegistration
class MappingAlgorithmFactory
class MissionControlFactory
class MappingAlgorithmDependencies
class MissionControlDependencies
class MatrixCell
class LoadedAlgorithmPlugin
class LoadedMissionControlPlugin

class PluginMatrixBinding {
    +plugin_filename

```

```

    +factory
}

class PluginMatrixResult {
    +plugin_filename
    +results
}

class IRunErrorLog {
    +log(ErrorRef)
}

class RunErrorLog {
    +log(ErrorRef)
}

class SimulationCoordUtil {
    <<module>>
    +worldInitialDronePosition()
    +forEachSphereSample()
}

class WavefrontPlanner
class MappingAlgorithmFrontier
class PathShaping {
    <<module>>
}
class ScanPlanning {
    <<module>>
}
class ExplorationPlan
class ConeTemplateCache
class VoxelStamp

class SimulationRunFactoryImpl {
    +create(...) ISimulationRun
}

class SimulationRunImpl {
    +run() SimulationResult
}

class SimulationImpl {
    +run(composition, output_path) SimulationManagerReport
}
class ISimulation

```

```

class Map3DImpl
class MockGPS
class MockLidar
class MockMovement

class MissionControlImpl_207190406_209543255 {
    +runMission() MissionRunResult
}

class DroneControlImpl {
    +step() DroneStepResult
    +state() DroneState
}

class MappingAlgorithmImpl_207190406_209543255 {
    +nextStep(state, scan) MappingStepCommand
}

class IMissionControl
class IDroneControl
class IMappingAlgorithm
class IMap3D
class IMutableMap3D
class ILidar
class IGPS
class IDroneMovement
class ISimulationRun
class ISimulationRunFactory

main --> SimulationCli
main --> SimulatorPaths
main --> YamlConfigParsers
main --> PluginLoader
main --> runPluginMatrix
main --> SimulatorReports
runPluginMatrix --> distributeWork
runPluginMatrix --> SimulationRunFactoryImpl
runPluginMatrix --> MatrixCell : expandRunMatrix
runPluginMatrix --> PluginMatrixBinding
runPluginMatrix --> PluginMatrixResult
PluginMatrixBinding --> ISimulationRunFactory
distributeWork --> SimulationRunFactoryImpl
SimulationRunFactoryImpl ..|> ISimulationRunFactory
SimulationRunFactoryImpl --> SimulationRunImpl
SimulationRunFactoryImpl --> MappingAlgorithmFactory

```

```

SimulationRunFactoryImpl --> MissionControlFactory
SimulationRunFactoryImpl --> MappingAlgorithmDependencies
SimulationRunFactoryImpl --> MissionControlDependencies
SimulationRunImpl ..|> ISimulationRun
SimulationRunImpl --> Map3DImpl
SimulationRunImpl --> MockGPS
SimulationRunImpl --> MockLidar
SimulationRunImpl --> MockMovement
SimulationRunImpl --> compareMaps
SimulationRunImpl --> RunErrorLog
SimulationRunImpl --> IMissionControl
SimulationRunImpl --> IMappingAlgorithm
SimulationImpl ..|> ISimulation
SimulationImpl --> runPluginMatrix
SimulationImpl --> ISimulationRunFactory
PluginLoader --> DlHandle
PluginLoader --> LoadedAlgorithmPlugin
PluginLoader --> LoadedMissionControlPlugin
PluginLoader ..> PluginRegistrar : registration ctors
MappingAlgorithmRegistration --> PluginRegistrar : setPendingAlgorithmFactory
MissionControlRegistration --> PluginRegistrar : setPendingMissionControlFactory
LoadedAlgorithmPlugin --> MappingAlgorithmFactory
LoadedMissionControlPlugin --> MissionControlFactory
MappingAlgorithmFactory --> MappingAlgorithmDependencies
MissionControlFactory --> MissionControlDependencies
MissionControlImpl_207190406_209543255 ..|> IMissionControl
MissionControlImpl_207190406_209543255 --> DroneControlImpl
DroneControlImpl ..|> IDroneControl
DroneControlImpl --> IMappingAlgorithm
DroneControlImpl --> ILidar
DroneControlImpl --> IGPS
DroneControlImpl --> IDroneMovement
DroneControlImpl --> applyScanToMap
applyScanToMap --> IMutableMap3D
MappingAlgorithmImpl_207190406_209543255 ..|> IMappingAlgorithm
MappingAlgorithmImpl_207190406_209543255 --> IMap3D
MappingAlgorithmImpl_207190406_209543255 --> WavefrontPlanner
MappingAlgorithmImpl_207190406_209543255 --> ExplorationPlan
MappingAlgorithmImpl_207190406_209543255 --> ScanPlanning
MappingAlgorithmImpl_207190406_209543255 --> ConeTemplateCache
MappingAlgorithmImpl_207190406_209543255 --> VoxelStamp
WavefrontPlanner *-- MappingAlgorithmFrontier : frontier_
WavefrontPlanner --> PathShaping
WavefrontPlanner --> ScanPlanning
WavefrontPlanner --> ExplorationPlan
Map3DImpl ..|> IMutableMap3D

```

```
ImmutableMap3D --|> IMap3D
MockLidar ..|> ILidar
MockGPS ..|> IGPS
MockMovement ..|> IDroneMovement
RunErrorLog ..|> IRunErrorLog
SimulationCoordUtil ..> Map3DImpl : world spawn
SimulatorPaths ..> RunErrorLog : errorLogPathFromOutputMap
YamlConfigParsers ..> IRunErrorLog
```

Sequence: one comparative cell

Comparative mode fixes one algorithm `.so` and varies every `MissionControl` `.so` in a folder. `main` parses CLI, creates the output dir, parses the composition (nested `parseSimulationConfig` / `parseMissionConfig` / `parseDroneConfig` / `parseLidarConfig`), loads plugins, takes pending factories, and builds one `SimulationRunFactoryImpl` / `PluginMatrixBinding` per plugin. `runPluginMatrix` expands the full cell list once, then one `distributeWork` call covers the whole plugin $\times$ cell matrix. Per-run `RunErrorLog` is opened from `errorLogPathFromOutputMap` before `runMission()`; startup errors skip the mission and score `kErrorScore`. After `runMission()` the output map is saved before `compareMaps`. Reports are `writeSimulationOutputYaml` per plugin, then `writeModeReport` / `writeComparativeReport`.

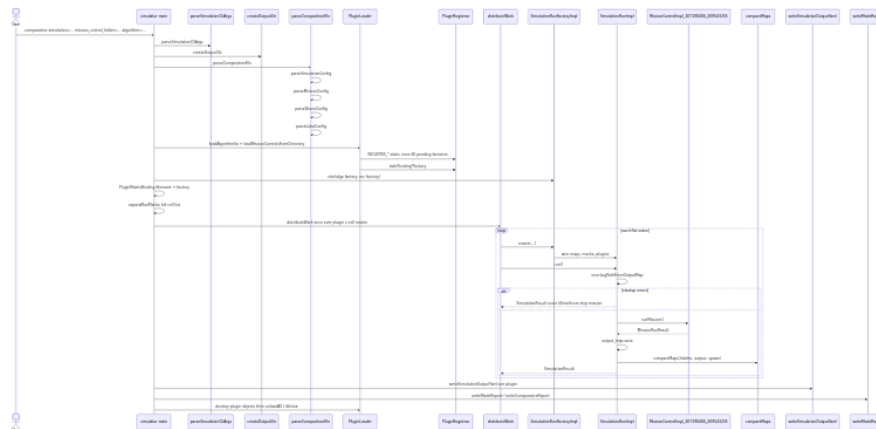


Figure 2: Comparative cell sequence

```
sequenceDiagram
    actor User
    participant Main as simulator main
    participant Cli as parseSimulationCliArgs
    participant Out as createOutputDir
    participant Comp as parseCompositionFile
    participant Loader as PluginLoader
```

```

participant Reg as PluginRegistrar
participant Dist as distributeWork
participant Factory as SimulationRunFactoryImpl
participant Run as SimulationRunImpl
participant MC as MissionControlImpl_207190406_209543255
participant Score as compareMaps
participant Yaml as writeSimulationOutputYaml
participant Rep as writeModeReport

```

```

User->>Main: -comparative simulation=... mission_control_folder=... algorithm=...
Main->>Cli: parseSimulationCliArgs
Main->>Out: createOutputDir
Main->>Comp: parseCompositionFile
Comp->>Comp: parseSimulationConfig
Comp->>Comp: parseMissionConfig
Comp->>Comp: parseDroneConfig
Comp->>Comp: parseLidarConfig
Main->>Loader: loadAlgorithmSo + loadMissionControlsFromDirectory
Loader->>Reg: REGISTER_* static ctors fill pending factories
Loader->>Reg: takePending*Factory
Main->>Factory: ctor(algo factory, mc factory)
Main->>Main: PluginMatrixBinding filename + factory
Main->>Main: expandRunMatrix full cell list
Main->>Dist: distributeWork once over plugin x cell matrix
loop each flat index
    Dist->>Factory: create(...)
    Factory->>Run: wire maps, mocks, plugins
    Dist->>Run: run()
    Run->>Run: errorLogPathFromOutputMap
    alt startup errors
        Run-->>Dist: SimulationResult score kErrorScore skip mission
    else
        Run->>MC: runMission()
        MC-->>Run: MissionRunResult
        Run->>Run: output_map.save
        Run->>Score: compareMaps(hidden, output, spawn)
        Run-->>Dist: SimulationResult
    end
end
Main->>Yaml: writeSimulationOutputYaml per plugin
Main->>Rep: writeModeReport / writeComparativeReport
Main->>Loader: destroy plugin objects then unloadAll / dlclose

```

Competition mode is the mirror image: one MissionControl .so is fixed and every algorithm in a folder is varied; the same runPluginMatrix / factory / run path applies (writeCompetitiveReport instead of writeComparativeReport).

Sequence: DroneControl step loop

Inside `MissionControlImpl_207190406_209543255::runMission()`, each iteration calls `DroneControlImpl::step()`. Each step carves the drone footprint, then invokes `nextStep` once unless draining a split-oversize queue. An unsupported movement type is retried inside `step()` up to `kMaxInvalidCommandRetries` before any Movement call; after N failures `step()` throws (`SimulationRunImpl` maps that to `MISSION_EXCEPTION`). `AlgorithmStatus::Finished` returns `Completed` with no move or scan. Over-size Advance/Elevate/Rotate is split into in-limit fragments (`splitWithinLimits`); the first fragment runs on the `nextStep` step and the rest drain from `pending_movements_` with no further `nextStep` and no extra scan — an intentional exception to “one `nextStep` per `step()`”, only when the algorithm exceeded drone max. Otherwise an optional movement runs, then at most one scan if the command carries `scan_orientation` (including after a recoverable `Continue`), matching the published movement → scan → fuse contract. Advance/Elevate are clamped to `mission_bounds` when those bounds are set; if the predicted center is still outside the world/map (`output_map_.isInBounds`), the command is ignored and Movement is not called (clamp then ignore). A step Error is pushed as `DRONE_STEP_FAILED` and the loop continues; it does not set `MissionRunStatus::Error` by itself.

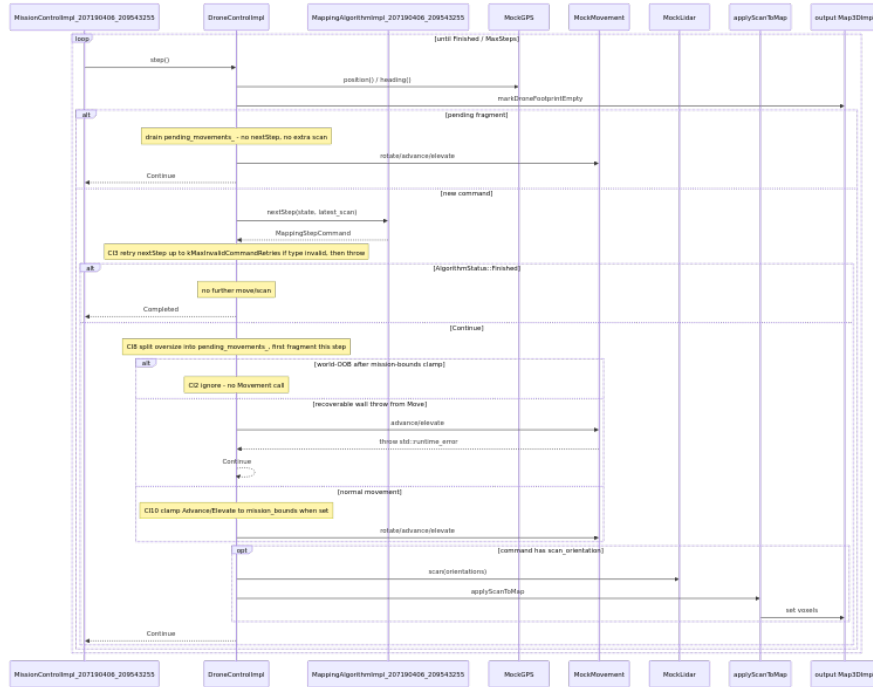


Figure 3: Drone step sequence

```

sequenceDiagram
    participant MC as MissionControlImpl_207190406_209543255
    participant DC as DroneControlImpl
    participant Algo as MappingAlgorithmImpl_207190406_209543255
    participant GPS as MockGPS
    participant Move as MockMovement
    participant Lidar as MockLidar
    participant SR as applyScanToMap
    participant Map as output Map3DImpl

    loop until Finished / MaxSteps
        MC->>DC: step()
        DC->>GPS: position() / heading()
        DC->>Map: markDroneFootprintEmpty
        alt pending fragment
            Note over DC: drain pending_movements_ - no nextStep, no extra scan
            DC->>Move: rotate/advance/elevate
            DC-->>MC: Continue
        else new command
            DC->>Algo: nextStep(state, latest_scan)
            Algo-->>DC: MappingStepCommand
            Note over DC: CI3 retry nextStep up to kMaxInvalidCommandRetries if type invalid
            alt AlgorithmStatus::Finished
                Note over DC: no further move/scan
                DC-->>MC: Completed
            else Continue
                Note over DC: CI8 split oversize into pending_movements_, first fragment th
                alt world-OOB after mission-bounds clamp
                    Note over DC: CI2 ignore - no Movement call
                else recoverable wall throw from Move
                    DC->>Move: advance/elevate
                    Move-->>DC: throw std::runtime_error
                    DC-->>DC: Continue
                else normal movement
                    Note over DC: CI10 clamp Advance/Elevate to mission_bounds when set
                    DC->>Move: rotate/advance/elevate
                end
                opt command has scan_orientation
                    DC->>Lidar: scan(orientations)
                    DC->>SR: applyScanToMap
                    SR->>Map: set voxels
                end
                DC-->>MC: Continue
            end
        end
    end
end

```

Recoverable collision handling: `MockMovement::advance / elevate` detect wall/boundary collisions against the hidden map and **throw** `std::runtime_error` (they never return a failed `MovementResult` for that case). Pre-throw limit checks in `MockMovement` return `{false, message}` when a command exceeds `max_rotate / max_advance / max_elevate`. `DroneControlImpl` catches the exception and any failed `MovementResult` and returns `DroneStepStatus::Continue`. `applyScanIfRequested` still runs after that `Continue` when the command carried a `scan_orientation`.

Backstop at the run boundary: Non-recoverable exceptions propagate out of `DroneControlImpl::step()` through `runMission()`. `SimulationRunImpl::run()` wraps the entire `runMission()` call in `try/catch`, logs the error, still saves the partial output map when possible, assigns score `kErrorScore` (-1), and lets the run matrix continue.

Threading

CLI	Behavior
<code>num_threads</code> absent	Main thread runs all cells
<code>num_threads=1</code>	Main thread runs all cells
<code>num_threads=N</code> ($N \geq 2$)	Up to N worker threads plus the main thread (workers capped at cell count)

All `.so` files are loaded on the main thread before workers start. Each matrix cell creates fresh plugin instances via the stored factories — instances are never shared or cached between runs. The result table is pre-allocated; workers write disjoint indices with no mutex on the table itself. Aggregate YAML reports are written on the main thread after all workers join. `.so` handles are not `dlclose`d from worker threads.

Data and maps

Each run owns two `Map3DImpl` instances:

- **Hidden map** — Loaded from the simulation config `.npz`. Wired into `MockLidar` and `MockMovement` only. Never exposed to plugins.
- **Output map** — Created empty (or from mission resolution settings). Passed to `DroneControlImpl` as `IMutableMap3D`; mission control writes lidar fusion here.

The algorithm receives `const common::IMap3D&` through `MappingAlgorithmDependencies` and reads voxel occupancy for planning only — it never mutates the map. All scan writes go through `DroneControlImpl` → `applyScanToMap` → output `Map3DImpl`.

After the mission, `SimulationRunImpl` saves the output map then calls `compareMaps(hidden, output, spawn)` to score it. World $\leftrightarrow$ voxel conversion and axis offsets are centralized in `user_common_207190406_209543255::SimulationCoordUtil(worldInitialDronePosition, forEachSphereSample)`.